

RDMA over Ethernet Investigation: Performance, Reliability, and Cost Analysis

Network Infrastructure Research

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This investigation examines RDMA over Ethernet (RoCE) as an alternative to proprietary solutions like InfiniBand for high-performance computing and AI training workloads. Based on Meta's large-scale deployment experience and comprehensive industry analysis, RoCE demonstrates competitive performance at significantly lower cost while maintaining good reliability for distributed AI training.

Executive Summary

This study analyzes three critical aspects of RDMA over Ethernet deployment: performance characteristics, reliability considerations, and economic trade-offs between standards-based and proprietary solutions.

Key Findings:

- **Performance:** RoCE achieves approximately 90-95% of InfiniBand performance in real-world AI workloads
- **Reliability:** Requires careful engineering but achieves production-grade reliability
- **Cost:** 55% lower Total Cost of Ownership (TCO) compared to InfiniBand solutions

Key findings demonstrate RoCE achieving approximately 90-95% of InfiniBand performance in real-world AI workloads with 55% lower Total Cost of Ownership.

Performance Analysis

Latency Characteristics

Meta's production deployment reveals significant insights into RoCE latency performance [Gangidi et al., 2024]. The company observed higher unloaded RTT (22 microseconds) due to CTSW switches using Virtual Output Queuing (VOQ) architecture, requiring careful tuning of collective library parameters.

Critical latency measurements include:

- Port-to-port latency: Ethernet switches 230ns vs InfiniBand 100ns
- End-to-end latency: InfiniBand 3-5 microseconds vs Ethernet 20-80 microseconds
- Production RTT observed: 22 microseconds for Meta's deployment

InfiniBand exhibits 10% performance improvement over RoCE for small 8-byte messages, but this gap narrows significantly for larger AI workloads.

Bandwidth and Throughput

Current bandwidth capabilities show Ethernet gaining competitive advantage:

Technology	Current Max	Port Speed
InfiniBand	51.2 Tb/s	400 Gb/s
Ethernet	51.2 Tb/s	800 Gb/s

Table 1: Bandwidth comparison between technologies

Meta’s optimization results demonstrate substantial performance gains:

- Enhanced ECMP with QP scaling: 40% improvement for AllReduce operations
- Overall optimization: greater than 2x improvement over baseline RoCE performance
- Collective-specific tuning: Critical for achieving production performance

Meta achieved more than 2x improvement in out-of-the-box RoCE performance through co-tuning network and collective library configurations.

AI-Specific Performance Considerations

Collective communication patterns significantly impact network performance. Meta’s analysis reveals:

Collective Type	Buffer Occupancy	Avg QPs per GPU
AlltoAll(v)	65.6 MB	15
AllReduce	13 MB	4
AllGather	22.1 MB	4
ReduceScatter	19.6 MB	4

Table 2: Traffic statistics in production (128 GPU configuration)

Reliability Analysis

Packet Loss and Error Handling

The fundamental reliability difference between technologies centers on loss tolerance:

InfiniBand Approach:

- Credit-based flow control guarantees lossless communication
- Hardware-level reliability mechanisms
- Native congestion avoidance

RoCE Requirements:

InfiniBand provides lossless fabric through credit-based flow control, while Ethernet requires careful engineering to achieve similar reliability.

- Lossless Ethernet configuration via Priority Flow Control (PFC)
- DSCP-based traffic classification
- Go-back-N retransmission for error recovery

Meta's Reliability Solutions

Meta's production deployment implements comprehensive reliability mechanisms [Gangidi et al., 2024]:

Flow Control Strategy:

- Single lossless queue across all switches and NICs
- DSCP-based PFC implementation
- Deep buffer allocation in spine switches (CTSWs)

Monitoring and Detection:

- Out-of-Sequence (OOS) packet tracking
- Link Flap Counter monitoring
- Local ACK Timeout (LAT) measurement
- PFC Watchdog for deadlock detection (greater than 200ms threshold)

Congestion Management Evolution

Meta's experience reveals important insights about congestion control effectiveness:

DCQCN Limitations:

- Challenging to tune for collective communication patterns
- Marginal benefits (3%) came with 2-3x worse PFC metrics
- Firmware bugs and reduced visibility in 400G deployments

Receiver-Driven Approach:

- Collective library controls traffic admission
- High priority queuing for CTS packets
- Superior performance compared to traditional congestion control

Meta operated 400G deployments with only PFC for flow control, demonstrating that transport-level congestion control may not be necessary for all AI workloads.

Cost Analysis: Proprietary vs Standards

Total Cost of Ownership Comparison

Juniper’s comprehensive TCO analysis provides detailed economic comparison between InfiniBand and RoCE deployments [Fetterolf, 2024].

Component	InfiniBand	Juniper Ethernet
CapEx	\$11.67M	\$5.29M
OpEx (3-year)	\$8.12M	\$3.55M
Total TCO	\$19.75M	\$8.84M
TCO Savings	\$10.91M (55%)	

Table 3: Three-year TCO comparison for 1024 GPU configuration

InfiniBand switches cost approximately double that of Ethernet switches, primarily due to sole-sourcing vs competitive industry production.

Economic Drivers

CapEx Advantages:

- InfiniBand hardware sole-sourced vs competitive Ethernet market
- Ethernet switches approximately 50% cost of equivalent InfiniBand
- Broader vendor ecosystem drives price competition

OpEx Benefits:

- Ethernet operational familiarity reduces training costs
- Automation tools (e.g., Juniper Apstra) provide 56% OpEx savings
- Reduced troubleshooting and outage remediation costs

Vendor Lock-in Considerations

InfiniBand Limitations:

- Proprietary nature restricts deployment flexibility
- Specialized skills requirement increases operational costs
- Vendor-dependent upgrade cycles limit flexibility

RoCE Advantages:

- Multi-vendor support: Mellanox, Xilinx, Broadcom, Intel
- Standards-based approach enables competitive sourcing
- Compatibility with existing Ethernet infrastructure

Meta's Production Experience

Architecture and Scale

Meta's deployment represents one of the largest production RoCE implementations for AI training:

Scale Achievements:

- Multiple clusters supporting thousands of GPUs each
- Maximum deployment: 32,000 GPUs per cluster
- Llama3 training: 16,000 GPUs on 24,000 GPU cluster

Network Architecture:

- Dedicated backend network separation
- Non-blocking Clos topology
- RoCEv2 protocol implementation

Meta successfully trained Llama3 on 16,000 GPUs using RoCE infrastructure, demonstrating viability for the most demanding AI workloads.

Engineering Optimizations

Co-tuning Requirements: Meta discovered that NCCL delivered sub-optimal out-of-box performance, requiring extensive tuning:

- Channel buffer size optimization for higher RTT
- Message size tuning for collective operations
- Clear-to-Send (CTS) packet prioritization
- Network topology-aware rank assignment

Routing Evolution: Meta's routing strategy evolved through multiple generations:

1. ECMP (inadequate for low-entropy traffic)
2. Path Pinning (fragmentation issues)
3. Enhanced ECMP with QP Scaling (40% improvement)
4. Centralized Traffic Engineering (optimal but complex)
5. Future: Flowlet Switching (promising for operational simplicity)

Industry Perspective and Future Outlook

Technology Convergence

Recent industry analysis indicates significant convergence between Ethernet and InfiniBand performance characteristics [WWT, 2025].

Performance Gap Narrowing:

- Ultra-high-performance domain (top 3-5%) still favors InfiniBand
- Vast majority of InfiniBand deployments addressable by Ethernet
- Advanced Ethernet features approaching InfiniBand capabilities

Emerging Solutions:

- NVIDIA SpectrumX: InfiniBand-like performance using 400Gb/s RoCE
- Broadcom Jericho3-AI: DPU-managed congestion control
- Ultra Ethernet Consortium: Standards-based high-performance networking

WWT's controlled testing using MLPerf benchmarks showed statistically insignificant performance delta (less than 0.03%) between InfiniBand and Ethernet for generative AI tests.

Market Dynamics

Current market trends strongly favor Ethernet adoption [FS Community, 2024]:

- Current AI deployments: approximately 90% InfiniBand in less than 10% of total switch market
- Projected 2025: 90% of AI workloads on Ethernet
- Market size: \$20 billion AI networking market

Recommendations

Decision Framework

Choose RoCE when:

- Avoiding vendor lock-in is priority
- Integration with existing Ethernet infrastructure required
- 5-10% performance trade-off acceptable for cost savings
- Operational complexity and automation matter
- Large-scale deployments where TCO is critical

Consider InfiniBand for:

- Ultra-low latency requirements (less than 3 microseconds)
- Specialized HPC workloads requiring minimum jitter
- Existing InfiniBand expertise and infrastructure
- Applications where every microsecond matters

*Implementation Best Practices***For RoCE Deployments:**

1. **Co-tune network and application layers** - Baseline performance insufficient
2. **Implement proper flow control** - PFC and receiver-driven admission critical
3. **Invest in automation** - Operational complexity requires tooling
4. **Design for reliability** - Redundancy and error handling essential
5. **Ensure expertise** - Both Ethernet and RDMA knowledge required

Conclusion

RDMA over Ethernet represents a mature, cost-effective alternative to proprietary solutions for most AI and HPC workloads. While InfiniBand maintains performance advantages in ultra-low latency scenarios, RoCE delivers 90-95% of the performance at 45% of the cost when properly engineered.

Meta's production experience at 24,000+ GPU scale validates that standards-based solutions can compete with proprietary alternatives. The combination of lower costs, vendor choice, operational familiarity, and rapidly improving performance positions RoCE favorably for most organizations building AI infrastructure.

The industry trajectory clearly favors Ethernet dominance in AI networking. Organizations should evaluate their specific latency requirements, cost constraints, and operational capabilities when choosing between these technologies, with most finding RoCE meets their needs at substantially lower cost.

Meta's successful deployment demonstrates that Ethernet-based RDMA can meet demanding requirements of large-scale AI training when properly co-tuned.

*Frequently Asked Questions**What is low-entropy traffic?*

Low-entropy traffic refers to network flows with limited diversity in their characteristics, particularly in the UDP 5-tuple (source IP,

destination IP, source port, destination port, protocol). This creates challenges for traditional load balancing mechanisms like Equal-Cost Multi-Path (ECMP) routing.

Characteristics of Low-Entropy Traffic in AI Workloads:

- **Limited flow diversity:** Fewer unique flows compared to traditional data center workloads
- **Predictable patterns:** Collective communication creates repetitive traffic patterns
- **Sparse topologies:** Training jobs use specific GPU-to-GPU communication patterns
- **Hash collisions:** ECMP struggles with poor flow distribution due to limited entropy

Impact on Network Performance: Low-entropy traffic causes uneven load distribution across network paths. Meta observed Max-Mean Ratio (MMR) over 1.2 for 1,000 flows on 16 links in simulations, with much worse performance in production. This imbalance results in:

- Collision of large elephant flows on the same network paths
- Severe congestion on overloaded links while other links remain underutilized
- Degraded overall network throughput and job performance

Solutions for Low-Entropy Traffic:

1. **Enhanced ECMP:** Additional hashing on destination Queue Pair (QP) fields using switch UDF capabilities
2. **QP Scaling:** Increasing the number of Queue Pairs to create more flows for better distribution
3. **Centralized Traffic Engineering:** Explicit flow placement based on real-time topology and traffic matrix
4. **Flowlet Switching:** Hardware-assisted load balancing that reacts to port load at microsecond granularity

Meta's experience demonstrates that addressing low-entropy traffic is critical for achieving optimal performance in AI training networks, requiring departure from traditional networking approaches designed for high-entropy web traffic.

Meta observed that AI training workloads typically have one GPU per training job with sparse logical topologies, resulting in predictable, repetitive flow patterns that confound hash-based load balancing.

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